



Entrepreneurial finance, credit cards, and race[☆]

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ABSTRACT

This paper examines the impact of financial deregulation on entrepreneurship. We assess the impact of credit card deregulation on transitions into self-employment using state-level removal of credit card interest rate ceilings following the US Supreme Court's 1978 *Marquette* decision as a quasi-natural experiment. We find that credit card deregulation increases the probability of entrepreneurial entry, with a particularly strong effect for black entrepreneurs. We demonstrate that these effects are magnified in states with a history of racial discrimination and link the results to discrimination-based barriers to entry.

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1. Introduction

This paper examines the impact of financial deregulation on entrepreneurship, a key driver of economic growth. We provide evidence that the deregulation of

credit card markets in the late 1970s expanded access to credit in the US economy, enabling liquidity-constrained individuals to borrow money and increase the rate of new businesses formation. While several previous studies of US financial deregulation investigate how commercial banking sector liberalization influenced economic growth through firm entry and exit, other less examined examples exist of financial deregulation that were significant enough to spur new firm formation. In particular, despite anecdotal evidence about the importance of credit cards in financing new enterprises, no previous study has explored how exogenous policy shocks to the availability of credit cards influences key economic activities, such as entrepreneurship.

Our empirical approach leverages differential credit constraints facing black and white entrepreneurs by estimating the impact of credit market deregulation on entrepreneurship by race. This strategy is underpinned by an important finding from previous studies that, depending on demographic characteristics, some individuals are more likely to use credit cards to finance their ventures

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than others. In particular, black entrepreneurs are more likely to finance their ventures using credit cards than white entrepreneurs due to differences in frictions encountered when accessing traditional bank loans and other external finance (Blanchflower, Levine, and Zimmerman, 2003; and Fairlie and Robb, 2008). Specifically, we use a differences-in-differences approach that exploits the removal of state level credit card interest rate ceilings following the US Supreme Court's 1978 decision in *Marquette National Bank of Minneapolis v. First Omaha Service Corp.* Our research design helps rule out plausible alternative explanations and establishes a credible causal link between credit card deregulation and entrepreneurial entry.

We first use data contemporaneous to the Supreme Court's *Marquette* decision to demonstrate that black borrowers were systematically more likely than white borrowers to face barriers to finance in the 1970s and 1980s. Our finding accords with results in Blanchflower, Levine, and Zimmerman (2003) who study barriers to finance using data from the 1990s. We then examine differences in credit card availability and ownership patterns following removal of state-level credit card interest rate ceilings. The patterns reveal that individuals based in states with no ceiling on credit card interest rates had more credit card debt and higher annual percentage rates (APRs) than individuals in states not affected by a similar policy change. These findings complement a study by Zinman (2002), which shows a significant increase in credit card ownership following a state's removal of its credit card interest rate ceiling, and are in line with anecdotal evidence that credit card issuers were likely to move to states without ceilings following a state-level policy change (Ausubel, 1997). More broadly, the results are consistent with the findings in Gross and Souleles (2002) linking credit card debt to changes in credit limits.

After establishing the link between a state's removal of credit card interest rate ceilings and the amount of credit cards in the state, we next examine how this type of credit card deregulation affected entrepreneurial entry. To do this we use data from the *Current Population Survey* (CPS) for 1971–1990 on transitions into self-employment. Our results suggest that living in a no-limit state resulted in a significant increase in the probability of a transition into self-employment, and the effect is particularly pronounced for black entrepreneurs. The results are robust to alternative models, including multinomial logit, and alternative specifications. Thus, one of the main contributions of our paper is to use a large sample setting to demonstrate the importance of credit cards to entrepreneurs. A likely explanation for the differential effect for black entrepreneurs is that, due to discrimination in traditional lending markets, black entrepreneurs with good projects relied more heavily on credit cards to fund new ventures than did white entrepreneurs. Such an explanation was originally suggested by Blanchflower, Levine, and Zimmerman (2003, p. 940), who write that “if financial institutions discriminate against blacks in obtaining small-business loans, we may even expect to see them use credit cards more often than whites, because they have fewer alternatives.”

To examine this explanation in more detail, we next test whether the effects of credit card deregulation on black entrepreneurial entry differ depending on the history of discrimination in the state. We split states along several measures of discrimination and show that the effect on black transitions into self-employment is larger in states with a history of discrimination. These results suggest that the increase in competition between credit card companies following a state's removal of its credit card interest rate ceiling reduced discrimination-based barriers to entry for black entrepreneurs.

Finally, we assess the extent to which credit card deregulation was endogenously determined by factors important to our study, such as the percent of self-employed or black individuals in a state. We find no evidence that the timing of credit card deregulation depended on these variables. We use data provided by Kroszner and Strahan (1999) to instead provide evidence that the timing of credit card deregulation was related to political economy variables. We also show patterns from the *Survey of Consumer Finance* (SCF) that suggest black entrepreneurs are more likely to own credit cards than white entrepreneurs in states that remove credit card interest rate ceilings.

We believe our findings provide a substantial contribution to two streams of literature. First, we contribute to a stream of literature that links the role of financial development to economic growth (Fazzari, Hubbard, and Peterson, 1988; Kaplan and Zingales, 1997; Levine, 2005). This literature has recently focused on the relation between bank deregulation and entrepreneurship (Black and Strahan, 2002; Cetorelli and Strahan, 2006; Bertrand, Schoar, and Thesmar, 2007; Huang, 2008; Kerr and Nanda, 2009) and shows that the removal of financial constraints increases entrepreneurial entry.¹ Our paper adds to this literature by studying a different source of financial deregulation, namely, removal of barriers to the access of credit cards. In doing so, we also demonstrate that financial deregulation can differentially affect entrepreneurs depending on demographic characteristics, such as race. This finding is relevant to the literature that examines the social effects of changes to credit market competition (Garmaise and Moskowitz, 2006). We also build on Levine, Levkov, and Rubinstein (2008), which shows a larger decrease in the black-white wage gap following bank deregulation in states with comparatively higher discrimination.

We also add to existing literature on entrepreneurial finance that focuses primarily on sources of finance such as bank loans and venture capital (Kortum and Lerner, 2000; Hsu, 2004; Zarutskie, 2006; Hochberg, Ljungqvist, and Lu, 2007; Hellmann, Lindsey, and Puri, 2008; Kerr and Nanda, 2009) and has only recently started to focus on alternative lending sources. To the best of our knowledge, the only other study on the link between credit cards and entrepreneurial finance is Scott (2010), which uses *Kauffman Firm Survey* data to show that a number of entrepreneurs use credit cards to start companies, although other studies have focused on the link between consumer debt and credit cards (e.g.; Gross and Souleles,

¹ For dissenting views, see Petersen and Rajan (2002) and Hurst and Lusardi (2004).

2002). Other examples of alternative lending sources include Morse (2011), which finds that access to payday loans helps alleviate unanticipated financial distress, and Ravina (2008) and Pope and Sydnor (2011), which study online lending markets. Our findings are also related to Benmelech and Moskowitz 2010, whose study of historical interest rates in the US shows that tighter interest rate ceilings lower economic activity, particularly for small firms.

The remainder of the paper proceeds as follows. Section 2 provides background for our study. Section 3 describes our methods and data. Section 4 describes the main results. Section 5 concludes and discusses the implications of our analysis.

2. Background

This section provides background on several facets of our study. We first document a link between race and liquidity constraints. We then describe the US Supreme Court's *Marquette* decision and the effect that the decision had on state-level policies and credit card availability and use

2.1. Race and liquidity constraints

Prior literature using data from the 1990s to the present (Blanchflower, Levine, and Zimmerman, 2003; and Robb, Fairlie, and Robinson, 2009) shows that blacks are more likely than whites to be turned down by bank lenders. In results reported in Table 1, we verify that blacks were more likely than whites to be turned down, or fear being turned down, by bank lenders in the late 1970s and early 1980s. To do this, we report the correlations between survey respondents who self-identify as *black* and answers to selected questions from the 1977 and 1983 *Survey of Consumer Finances*, controlling for individual characteristics and state of residence.

The questions differ across the two surveys. For the 1977 survey, respondents were asked about their opinions on institutions that lend money or extend credit, including stores, banks, finance companies, and credit unions. Respondents were not asked to distinguish between lenders and creditors.² In Column 1, we report results of answers to the question: "In your opinion, have you ever been treated unfairly in your credit transactions?" In Column 2, we report results of answers to the question: "Are there any (other) practices of creditors or lenders that you think are unfair and would like to see changed?" For the 1983 survey, respondents were asked about their experience obtaining loans or credit. In Column 3, we report results of answers to the question: "In the past few years, has a particular lender or creditor turned down any request you (or your husband/wife) made for credit or have you been unable to get as much credit as you applied for?" In Column 4, we report results of answers to the question: "Was there any time in the

Table 1

Survey of Consumer Fairness (SCF) questions on fairness of lenders and availability of loans.

This table reports results of linear probability regressions of "Yes" answers to the questions indicated in each column on an indicator for black. The data are from the *Survey of Consumer Finances* for year indicated. Individual characteristics include female, age, high school graduate, married, homeowner, and household income. State fixed effects are included for 35 states covered by the SCF; the SCF excludes DC, HI, ID, KS, MD, MT, ND, NH, NM, NV, RI, VT, WV, and WY. Robust standard errors are included in brackets and clustered at state. * Significant at 10%; ** Significant at 5%; *** Significant at 1%.

	Treated unfairly? (1)	Unfair practices you want to change? (2)	Turned down or unable to obtain? (3)	Afraid of being turned down? (4)
Black	3.7915 [2.2671]	2.1454 [1.5042]	0.0692* [0.0370]	0.1155*** [0.0221]
Year	1977	1977	1983	1983
Individual characteristics	Y	Y	Y	Y
State fixed effects	Y	Y	Y	Y
Number of observations	1534	1534	2077	2080
R-squared	0.032	0.047	0.090	0.071
Clusters	35	35	35	35

past few years that you (or your husband/wife) *thought* of applying for credit at a particular place, but changed your mind because you thought you might be turned down?" (Emphasis in the original SCF survey questionnaire). Black individuals were more likely to answer yes to all four questions, and this result is statistically significant at the 10% level in Column 3 and 1% level in Column 4. Taken together, survey answers suggest that black individuals in the 1970s and 1980s encountered frictions, or believed they would encounter frictions, when attempting to access finance. Blanchflower, Levine, and Zimmerman (2003), using *Survey of Small Business Finance* data from 1993 and 1998, report qualitatively similar findings: Black-owned firms were more likely to report being concerned about credit market problems and less likely to apply for credit because of fear of being turned down.

2.2. State policy changes following the *Marquette* decision

In December 1978, the Supreme Court considered the case of *Marquette National Bank of Minneapolis v. First Omaha Service Corp.* The case centered around First Omaha's marketing of credit cards to customers in Minnesota. At that time, states were allowed to set their own caps on credit card interest rates, and the ceilings in Nebraska and Minnesota were different. Thus, First Omaha could charge a higher interest rate, as allowed by Nebraska law, than Minnesota-based banks could legally offer to customers in Minnesota. As a result, the Minnesota attorney general argued that First Omaha's activities interfered with the state's ability to enforce its usury laws. After a favorable state trial court decision for *Marquette* was overturned by the Minnesota Supreme Court, the case went to the US

² The specific language is: "In this interview please think of the terms 'creditors' and 'lenders' as the same thing."

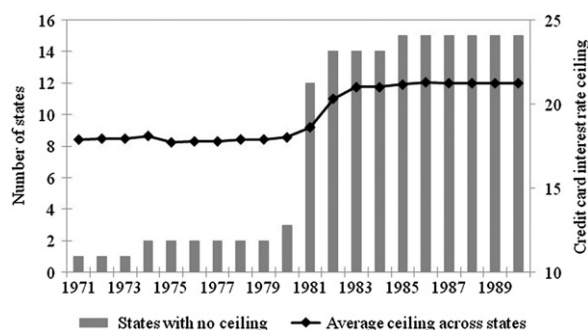


Fig. 1. The figure shows the increase in number of states with no ceiling on credit card interest rates following the Supreme Court's *Marquette* decision in December 1978 (left axis), as well as the average interest rate ceiling across states (right axis). By 1985, 15 states had no ceiling on credit card interest rates, up from one (New Hampshire) in 1971. States that removed caps during this time period were Arizona (1980), Delaware (1981), Idaho (1983), Illinois (1981), Montana (1981), Nevada (1981), New Jersey (1981), New Mexico (1981), North Dakota (1985), Oregon (1974), South Dakota (1981), Utah (1981), Virginia (1983), and Wisconsin (1981). Source of the data is *The Cost of Personal Borrowing in the United States, various years*.

Supreme Court.³ The Court ruled that the National Bank Act stipulated that nationally chartered banks could charge the highest allowable rate in their home state, regardless of the interest rate ceiling in the customer's state of residence (Ausubel, 1997). Two years later, the Depository Institutions Deregulation and Monetary Control Act became law, providing state chartered banks with similar protection to "export" interest rates across state boundaries (Furletti, 2004).

From 1980 to 1985 a number of states removed their credit card interest rate ceilings (see Fig. 1; New Hampshire and Oregon had no ceiling during this period). By removing the rate ceilings, these states switch from having a limit on credit card interest rates to having no limit. According to some accounts, states removed interest rate ceilings in an attempt to attract and retain banks, and major banks such as Citibank moved to no-limit states such as South Dakota and Delaware (DeMuth, 1986). However, despite Citibank's high profile move to South Dakota, there was not an immediate migration to no-limit states because of legal restrictions on interstate banking. Many of these restrictions remained in place until the mid-1980s (Kroszner and Strahan, 1999). As a result, there was not an immediate saturation of interstate credit cards marketed from banks in no-limit states to individuals in states with limits. Instead, individuals living in no-limit states were immediately affected, but not individuals residing in states with limits. Knittel and Stango (2003) report that, as of 1984, only 8–9% of customers held out-of-state bank cards.

2.3. Effect on credit card availability and use

Following their move to no-limit states, credit card companies significantly increased the marketing of their

cards. Marketing was primarily accomplished via direct mail solicitation. Accounts from the 1980s suggest that credit card companies aggressively and indiscriminately marketed their cards (DeMuth, 1986).⁴ Credit scoring technology, while available, was estimated to be used in only 20–30% of consumer credit decisions (Capon, 1982). Zinman (2002) uses *Survey of Consumer Finances* to study the effect of credit card deregulation and shows that credit card ownership significantly increased following a state's removal of credit card interest rate ceilings. In Table 2 we provide additional details on the effect of a state's removal of credit card interest rate ceilings on credit card supply. We first examine data from Knittel and Stango (2003) on state-level Herfindahl-Hirschman Index (HHI) of credit card companies. HHI is lower in states with no ceiling on credit card interest rates, but not statistically significant. Next, we use data from the *Survey of Consumer Finances* to study individual differences across states. By 1983, 72% of individuals living in limit states owned a credit card compared with 77% of individuals living in no-limit states. In addition, the distribution of financing provided by credit cards shifted to include higher interest rates and larger amounts of debt. The findings presented in Zinman (2002) and in Table 2 provide evidence that a state's switch to no limit increased the equilibrium quantity of credit cards and credit card debt in the state. Our empirical design takes advantage of this shock to examine the role of credit card availability on black and white entrepreneurial entry.

3. Empirical strategy and data

3.1. Empirical strategy

We hypothesize that access to credit cards is an important determinant of entrepreneurial activity. Our prediction is that the removal of state-level credit card interest rate ceilings following the *Marquette* decision increased entrepreneurship and that the effect was especially pronounced among blacks because of difficulty accessing traditional forms of external finance. We treat the state-level changes to credit card interest rate ceilings as exogenous deregulatory shocks to the availability of credit card financing. We subsequently provide empirical support that the shocks were exogenous with respect to the variables of interest in our study. We focus on transitions into self-employment as a measure of entrepreneurial entry. Accordingly, the main specification is

$$y_{imt}^* = \alpha + \beta_1 rate_{mt} + \beta_2 rate_{mt} * black_{imt} + \delta_m + T_t + trend * \delta_m + \mathbf{X}_{imt} \beta + e_{imt} \quad (1)$$

where y_{imt}^* is the probability of individual i living in market m transitioning from full-time paid employment to full-time self-employment at time t . When $y_{imt}^* > 0$, we observe $y_{imt} = 1$ indicating that the individual has

³ *Marquette National Bank of Minneapolis v. First of Omaha Service Corp.*, 439 US 299 (1978).

⁴ As one interesting example, an editor's footnote in DeMuth (1986) describes how several of the editors of *Yale Journal on Regulation* received mail solicitation for student-specific credit cards. See also Time (1986).

Table 2

Credit characteristics of states and individuals.

State-level data on Herfindahl-Hirschman Index (HHI) of credit card issuers are from Knittel and Stango (2003) and are available for 33 states in 1983 and 38 states in 1986. Individual-level credit data are from the *Survey of Consumer Finances*. The number of observations varies from 1,900 to 4,103, due to missing observations. APR=annual percentage rate.

	Sample restriction: no-limit state?		
	No	Yes	T-test
<i>State-level credit characteristics</i>			
State-level HHI credit card issuers (1983)	0.23	0.19	0.53
State-level HHI credit card issuers (1986)	0.27	0.17	1.40
<i>Individual-level credit characteristics</i>			
Credit card debt (1983)	283.58	370.35	−3.13
Last month's credit card balance (1983)	204.35	275.63	−2.89
APR on credit card (1983)	17.72	18.17	−2.09
Number of bank-issued credit cards (1983)	0.72	0.77	−1.32

transitioned to self-employment, and when $y_{imt}^* < 0$, the individual has not transitioned. $Rate_{mt}$ is the prevailing ceiling on credit card interest rates in the state. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year, an approach consistent with Benmelech and Moskowitz, 2010. The rate ceiling for no-limit states is 24% prior to 1981 and 25% in 1981 and after.⁵ Fig. 1 shows that, as more states switch to no limit, the average rate ceiling across states increases. We include year effects (T_t) to control for macroeconomic fluctuations that affect the employment opportunity set faced by each individual. We include market fixed effects (λ_m) to control for differences in employment opportunities, local regulations regarding business start-up costs, and other entry barriers across markets. A market is defined at the metropolitan statistical area (MSA)-state level. For example, the boundary of the Philadelphia metro area crosses into two states (Pennsylvania and New Jersey) and so is divided into two mutually exclusive areas. In addition, areas in each state not part of an MSA are grouped into a statewide non-MSA area. To allow for different trends across market areas we include an interaction between a time trend and the market fixed effect $Trend \times \delta_m$, an approach that follows Besley and Burgess (2004) and Wolfers (2006). X_{imt} is a vector of individual characteristics, including a dummy for *black*, and industry dummies.

Throughout all of our specifications the error terms e_{imt} are clustered at the MSA-state level to account for autocorrelation in the data across individuals. This clustering relaxes the assumption of independence of the error terms of individuals that live in close proximity to one another and ensures that the standard errors are not underestimated (Bertrand, Duflo, and Mullainathan, 2007). We are particularly interested in the coefficients β_1 and β_2 . We expect that a state's switch to no limit results in increased probability of transition into self-employment in the

population, with a particularly strong effect for black individuals. That is, we expect both β_1 and β_2 to be positive.

To test the role of credit cards as a mechanism that addresses discrimination-based barriers to entry, we categorize states along different measures of discrimination and compare β_2 across these state types. Specifically, we run Eq. (1) separately for different groups of states and then compare the resulting β_2 coefficients using χ^2 tests. For each measure of discrimination, we expect that $\beta_2^{high_discrimination} > \beta_2^{low_discrimination}$.

3.2. Description of data

Data on the credit card interest rate ceiling for each state during our sample period was hand-collected from annual volumes of *The Cost of Personal Borrowing in the United States*. We use *Current Population Survey* data from 1971 to 1990 to establish the link between changes in availability of credit card financing and self-employment transition rates. The CPS is ideal for this analysis because it includes many demographic variables that we use to control for alternative explanations. We restrict our observations to individuals who are white or black, who are between ages 18 and 65, who work full time, and who do not work for the military or on a farm. Consistent with other work in this area (e.g.; Fairlie, 1999), *transition into self-employment* is our dependent variable in all regressions on entrepreneurial entry. Self-employment is commonly used to identify entrepreneurs and is the best variable we have given the nature of the CPS data. For most models, we identify transitions into self-employment by restricting the sample to individuals who worked full time in paid employment in the prior year. The results are robust to including in the risk set workers in both paid employment and unemployment sectors.

We use a number of demographic characteristics from the CPS that previous studies have shown are important predictors of self-employment. These variables include indicators for *black*, *female*, *married*, *home owner*, *urban*, *high school graduate*, and *non-metro area* as well as continuous variables for *age* and its square. *Household income* is censored from above so we instead use a dummy to indicate if the household is in the bottom 20th percentile of household income in that year. It is particularly important to control

⁵ The results are consistent across several robustness checks suggested in Benmelech and Moskowitz 2010. The robustness checks include using a rate ceiling of 25% for no limit states across all years, using a rate ceiling of 30% for no limit states, and using a dummy variable equal to one when the state has no limit and zero otherwise. See Table A1 in the appendix.

Table 3

Summary statistics of variables used in analyses; full sample and split sample.

This table presents summary statistics of variables used in regressions. Data come from the *Current Population Survey (CPS)* for years 1971–1990. We restrict the sample to individuals who are white or black, who are between ages 18 and 65, who work full time, and who do not work for the military or on a farm. There are 546,612 observations. The variable *transition into self-employment* is constructed by limiting the sample to individuals who worked full time in the paid employment sector in the prior year who then switched into self-employment in the current year. The variable *rate* is the prevailing interest rate ceiling in the state, unless the state is a no-limit state in which case the rate is set equal to the highest rate ceiling across states in that year. *No limit* indicator equals one if the state has no interest rate ceiling and zero otherwise. For the split sample, a limit state is a state that never switches to no limit. A no-limit state is a state that switches to no limit by 1990 (the last year of the dataset). Data for the split sample are from 1977, which is the year prior to the Supreme Court's *Marquette* decision and the first year that the CPS includes information on all 50 states plus Washington, DC.

	Full sample				Split sample		
	Mean	Standard deviation	Minimum	Maximum	Limit	No limit	T-test
Transition into self-employment	0.009	0.097	0.000	1.000	0.007	0.008	−0.49
Black	0.086	0.281	0.000	1.000	0.088	0.066	6.33
Female	0.373	0.484	0.000	1.000	0.348	0.339	1.58
Homeowner	0.554	0.497	0.000	1.000	0.659	0.657	0.37
Household income	29,626	28,635	−10,618	999,999	18,788	19,483	−5.19
Age	37.020	12.207	18.000	65,000	37.0135	36.8777	0.85
High school graduate	0.786	0.410	0.000	1.000	0.734	0.764	−5.52
Married	0.659	0.474	0.000	1.000	0.699	0.708	−1.62
Non-metro	0.221	0.415	0.000	1.000	0.295	0.149	26.34
Unemployed % (percent)	0.033	0.013	0.000	0.167	0.037	0.036	5.61
Rural % (percent)	0.023	0.041	0.000	0.286	0.037	0.045	−13.88
Rate	0.196	0.030	0.100	0.250			
No limit indicator	0.120	0.325	0.000	1.000			

for low-income households given other research that shows changes in credit market competition affect income distribution (Beck, Levine, and Levkov, 2010) and business activity (Garmaise and Moskowitz, 2006). We also construct demographic variables by market for *unemployment rate* and *percent of population living in a rural area*. Self-employment transitions could vary by industry based on different financing needs across industries. For example, according to the Federal Reserve Board's 1987 *National Survey of Small Business Finance* (NSSBF), the median starting capital in the construction industry was \$9,500, whereas the median starting capital in retail trade was \$55,200.⁶ Hence, 67 industry dummies are included to control for differences in entrepreneurial entry rates across industry. The CPS data include weights, and the main results are robust to the use of these weights. However, consistent with the approach taken in Puri and Robinson (2009), we do not use weights in any of the reported results because our intent is to measure the effect of changes in availability of finance on an individual's decision to enter entrepreneurship.

Table 3 presents summary statistics and a comparison of variable means between states that removed their credit card interest rate ceiling (*no limit*) during the sample time frame and those that did not (*limit*). The comparison uses data from 1977 as that was the first year in which the CPS provided data from all 50 states and the District of Columbia and because 1977 is the year prior to the *Marquette* decision. Individuals living in no-limit states are less likely to be black, more likely to be high school graduates, more likely to have higher household income, and more likely to live in areas with lower unemployment and a higher percent of population in

rural areas. While there appear to be differences across the two types of states, our analyses rely on within-state effects. In addition, as reported in Section 4, we find no evidence that the percentages of self-employed, black, or black self-employed individuals in a state predict a state's hazard for removal of its credit card interest rate ceiling.

4. Results

This section reports our empirical results. We first provide results from our basic model and then show that the results are robust to alternative specifications. We also break the results out into split samples by various measures of discrimination. Finally, we discuss alternative explanations for our results, and provide further support for our findings.

4.1. Results from the basic model

Results of linear probability regressions are reported in all tables unless otherwise described; coefficients for control variables are suppressed for presentation purposes. Table 4 presents the results of entrepreneurial entry as described in Eq. (1) using *Current Population Survey* data from 1971 to 1990. Each column focuses on a different risk set of individuals. Column 1 investigates the effects of credit card availability on transitions into self-employment at time t from paid employment at time $t-1$. The coefficient on *rate* is 0.0300 and significant at the 5% level. The coefficient on *black*rate* is 0.0362 and significant at the 5% level. Column 2 next investigates the effects of credit card availability on transitions into self-employment at time t from unemployment at time $t-1$. None of the coefficients is significant. Finally, Column 3 investigates the effects of credit card availability on transitions into self-employment at time t from either

⁶ NSSBF statistics are cited in Hurst and Lusardi (2004). The earliest year for the NSSBF data is 1987.

Table 4

Effect of state level credit card interest rate ceilings on entrepreneurial entry.

This table reports linear probability models of transitions into self-employment on state-level credit card interest rate ceilings, black, and their interaction, using data from the *Current Population Survey* for 1971–1990. In Column 1 the variable transition into self-employment equals one if the individual worked in paid employment in the previous year but switched into self-employment in the current year and zero otherwise. In Column 2 the variable transition into self-employment equals one if the individual was unemployed in the previous year but switched into self-employment in the current year and zero otherwise. In Column 3 the variable transition into self-employment equals one if the individual was in paid employment or if the individual was unemployed in the previous year but switched into self-employment in the current year and zero otherwise. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year. This implies the rate ceiling varies from 24 for no-limit states prior to 1981 and 25 for no-limit states in 1981 and after. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, non-metro area indicator, local unemployment rate, and percent of local population living in rural areas. Fixed effects for years, 67 industries and 347 metropolitan statistical area (MSA)-state areas are included. Robust standard errors are included in brackets and clustered at the MSA-state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	Risk set		
	Paid employment in prior year (1)	Unemployment in prior year (2)	Paid employment or unemployment in prior year (3)
Rate	0.0300** [0.0125]	–0.117 [0.1446]	0.0209 [0.0134]
Black	–0.0087** [0.0035]	–0.0339 [0.0261]	–0.0098** [0.0038]
Black*rate	0.0362** [0.0177]	0.0488 [0.1284]	0.0332* [0.0189]
Individual characteristics	Y	Y	Y
Industry dummies	Y	Y	Y
Year fixed effects (1971–1990)	Y	Y	Y
MSA-state fixed effects	Y	Y	Y
Trend*MSA-state fixed effects	Y	Y	Y
Number of observations	546,612	28,014	574,626
R-squared	0.0158	0.1692	0.0218

paid employment or unemployment at time $t - 1$. The coefficient on *rate* is not significant, whereas the coefficient on *black*rate* is 0.0332 and significant at the 10% level. The coefficients on *black* are negative across all columns, indicating that black individuals are on average less likely than white individuals to transition into self-employment.

The effect of credit card deregulation appears to affect employment transitions of individuals in the paid employment sector but not unemployment sector. One reason could be the difference in characteristics of individuals that comprise these different risk sets. As shown in the Appendix, individuals in paid employment differ from individuals in unemployment along a number of dimensions (see Table A2). Hence, in subsequent models we restrict the sample to individuals who worked full time in paid employment in

the prior year, an approach that follows Evans and Jovanovic (1989), Holtz-Eakin, Joulfaian, and Rosen (1994), Fairlie (1999), and others.

4.2. Alternative models and specifications

In Table 5 we investigate the robustness of the results to alternative models. Column 1 presents results of a probit model on the choice to transition into self-employment. The coefficient on *black*rate* is positive and statistically significant. Columns 2 and 3 present results of a multinomial logit model of the choice between staying in the paid employment sector (the base case), transitioning into self-employment and transitioning into unemployment. In Columns 2 and 3, the coefficients in the models are relative to the base case. The coefficient on *black*rate*

Table 5

Effect of state-level credit card interest rate ceilings on entrepreneurial entry: alternative models of transition.

This table reports models of transitions into self-employment on state-level credit card interest rate ceilings, black, and their interaction, using data from the *Current Population Survey* for 1971–1990. The dependent variable *self-employment* equals one if the individual worked in paid employment in the previous year but switched into self-employment in the current year. The dependent variable *unemployment* equals one if the individual worked in paid employment in the previous year but switched into unemployment in the current year. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year. This implies the rate ceiling varies from 24 for no-limit states prior to 1981 and 25 for no-limit states in 1981 and after. Column 1 presents results of a probit model on the choice to transition into self-employment and includes the earnings difference to control for opportunity cost of the choice. Columns 2 and 3 present results of a multinomial logit model of the choice between staying in a full-time job (the base case), transitioning into self-employment (2) and transitioning into unemployment (3). The coefficients in models (2) and (3) are relative to the base case. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, non-metro area indicator, local unemployment rate and percent of local population living in rural areas. Fixed effects for years, 67 industries and metropolitan statistical area (MSA)-state areas are included. Robust standard errors are included in brackets and clustered at the MSA-state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	Probit model Self-employment (1)	Multinomial logit model	
		Self-employment (2)	Unemployment (3)
Rate	0.2277 [0.4868]	0.491 [1.2128]	–0.7212 [0.5152]
Black	–0.4807** [0.2002]	–1.2361** [0.5445]	0.2925* [0.1516]
Black*rate	1.9642* [1.0074]	5.2558* [2.7443]	1.4423* [0.8060]
Individual characteristics	Y	Y	Y
Industry dummies	Y	Y	Y
Year fixed effects (1971–1990)	Y	Y	Y
MSA-state fixed effects	Y	Y	Y
Number of observations	534,538	546,612	546,612
Pseudo R-squared	0.125	0.108	0.108

is positive and statistically significant in both columns, indicating that black individuals living in a state that raises or eliminates the ceiling on credit card interest rates is more likely to transition into self-employment and also more likely to transition into unemployment.

In Table 6 we present results from several robustness checks. Columns 1 and 2 investigate the sensitivity of the results to different year ranges. In Column 1 the year range is restricted to 1977–1990. The year 1977 was the first in which data on all states were reported, whereas prior to this period the CPS included data from a subset of US states. In Column 2 the year range is restricted to 1977–1985 so as to focus on the years immediately prior and following the *Marquette* decision. The coefficient on *black*rate* remains positive and significant across these different year ranges.

Columns 3–5 in Table 6 include various additional interaction terms to rule out plausible alternative explanations. Column 3 includes information on state bank branching deregulation. Banking deregulation was contemporaneous to credit card deregulation and so presents a potential confounding effect that could explain the results shown thus far. To address this possibility, we include an indicator for *interstate banking deregulation* and its interaction with *black*. The basic model includes only the interaction between *black* and *rate*, which might also capture unobserved interactions between *rate* and other indicators for low socioeconomic status that are correlated with *black*. Column 4 controls for this possibility by interacting *rate* with other individual characteristics. Column 5 includes interactions between *black* and the industry dummies to control for the possibility that black individuals

Table 6

Effect of state-level credit card interest rate ceilings on entrepreneurial entry, various specifications.

This table reports linear probability models of transitions into self-employment on state-level credit card interest rate ceilings, *black*, and their interaction, using data from the *Current Population Survey* for 1971–1990. The variable transition into self-employment equals one if the individual worked in paid employment in the previous year but switched into self-employment in the current year and zero otherwise. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year. This implies the rate ceiling varies from 24 for no-limit states prior to 1981 and 25 for no-limit states in 1981 and after. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, non-metro area indicator, local unemployment rate, and percent of local population living in rural areas. Column 1 restricts the sample to 1977–1990. Column 2 restricts the sample to 1977–1985. Column 3 uses the full sample and adds an indicator for interstate bank deregulation and its interaction with *black*. Column 4 uses the full sample and adds additional interactions between individual characteristics and the interest rate ceiling. Column 5 uses the full sample and adds additional interactions between *black* and industry dummies. Column 6 and Column 7 split the sample into low-cost and high-cost industries, respectively. Column 8 and Column 9 split the sample into those states with a low percent of national banks and high percent of national banks, respectively. Fixed effects for years, 67 industries and metropolitan statistical area (MSA)-state areas are included. Robust standard errors are included in brackets and clustered at the MSA-state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	Sample restriction								
	Year > 1976 (1)	Year > 1976 and Year < 1986 (2)	Full sample (3)	Full sample (4)	Full sample (5)	Low-cost industries (6)	High-cost industries (7)	Low percent national banks (8)	High percent national banks (9)
Rate	0.0281 [0.0180]	0.0185 [0.0193]	0.0031 [0.0272]	0.0050 [0.0276]	0.0302** [0.0125]	0.0293 [0.0361]	0.0442** [0.0187]	0.0242 [0.0255]	0.0365 [0.0298]
Black	−0.0100** [0.0041]	−0.0138*** [0.0046]	−0.0125*** [0.0032]	−0.0089** [0.0035]	−0.0190*** [0.0071]	−0.0177** [0.0072]	−0.0030 [0.0039]	−0.0135*** [0.0050]	−0.0064 [0.0053]
Black*rate	0.0441** [0.0202]	0.0726*** [0.0230]	0.0635*** [0.0166]	0.0372** [0.0172]	0.0328* [0.0174]	0.0862** [0.0362]	0.0088 [0.0200]	0.0596** [0.0251]	0.0282 [0.0263]
Individual characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry dummies	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
MSA-state fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Trend*MSA- state fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Bank deregulation interaction	–	–	Y	–	–	–	–	–	–
Rate*individual characteristics	–	–	–	Y	–	–	–	–	–
Black* industry dummies	–	–	–	–	Y	–	–	–	–
Number of observations	458,221	309,464	546,612	546,612	546,612	170,661	256,077	194,269	263,952
R-squared	0.0160	0.0153	0.0159	0.0159	0.0160	0.0186	0.0176	0.0178	0.0150

could be more likely to work in certain industries, perhaps due to different skills, preferences, or access to start-up capital. The coefficient on *black*rate* remains positive and significant across these different specifications.

Columns 6 and 7 in Table 6 investigate how *black*rate* varies by industry capital requirements. We use the NSSBF statistics cited in Hurst and Lusardi (2004) to designate an industry as low cost or high cost. The coefficient on *rate* is positive but not significant in Column 6 and positive and significant in Column 7, but a χ^2 test cannot reject the null hypothesis that the coefficients are the same across the two columns. The coefficient on *black*rate* is positive and significant in Column 6 and positive but not significant in Column 7. A χ^2 test rejects the null hypothesis that the coefficients are the same across the two Columns at the 10% level. These results suggest that the black entrepreneurial entry increased more in low capital intensive industries than in high capital intensive industries following removal of credit card interest rate ceilings.

Columns 8 and 9 in Table 6 investigate how *black*rate* varies by state-level bank composition. Under the *Marquette* decision, credit cards offered by nationally chartered banks are subject to the higher of the state-level credit card interest rate ceiling or the ceiling in the nationally chartered bank's home state. We, therefore, expect that the removal of a state's credit card interest rate ceiling has a larger effect in states with fewer nationally chartered banks. Prior to the removal of the ceiling, these states have fewer banks with the potential to offer credit cards at rates above the state's prevailing ceiling. We use Summary of Deposits data from the Federal Deposit Insurance Corporation (FDIC) to identify the share of deposits held by nationally chartered and state-chartered bank branches located within each state in 1980. We categorize a state as having a low percent of nationally chartered banks if its percent of nationally chartered banks is below the national median (results presented in Column 8), and otherwise categorize the state as high (results presented in Column 9). The coefficient on *rate* is positive but not significant in both columns. The coefficient on *black*rate* is positive and significant at the 5% level in Column 8 and positive but not significant in Column 9. However, a χ^2 test cannot reject the null hypothesis that the coefficients are the same across the two columns.

4.3. The role of discrimination

A consistent finding across the results in Tables 4–6 is that black individuals who reside in a state that increases the ceiling on credit card interest rates and who worked in the paid employment sector at $t-1$ were more likely to enter self-employment by time t . A likely explanation for the differential effect on black and white entrepreneurs is that black entrepreneurs faced discrimination in traditional lending markets. As a result, black entrepreneurs relied more heavily on credit cards to fund new ventures than did white entrepreneurs (Blanchflower, Levine, and Zimmerman, 2003). To understand the role of discrimination in access to credit, we investigate whether the impact of credit card deregulation differentially affected black

entrepreneurs in states with a history of discrimination. As argued in prior research, variation in institutions and norms in an earlier time period can explain variation across these same areas in later periods (Acemoglu, Johnson, and Robinson, 2001). Thus, we first focus on historical state characteristics by identifying states that allowed slavery at the start of the Civil War (*slave state*). We next focus on more recent state characteristics contemporaneous to the *Marquette* decision. We identify states that were among the last to remove anti-miscegenation laws (*anti-miscegenation law state*). We obtain information on the states that repealed anti-miscegenation laws after the US Supreme Court's 1967 decision in *Loving v. Virginia* from Fryer (2007). We also identify states that did not have fair housing laws (*no fair housing law state*) until the federal Fair Housing Act of 1968 from Collins (2004). Finally, we use the racial bias index reported in Levine, Levkov, and Rubinstein (2008), which measures the difference between actual and predicted interracial marriage rates in 1970, to classify states as above or below the median interracial marriage bias (*interracial marriage bias state*).

All the results presented in Table 7 replicate the model in Table 4, Column 1 with results split by state type across adjacent columns. Column 1 focuses on states that were not slave states immediately prior to the Civil War; the coefficient on *black*rate* is -0.0125 but not significant. Column 2 focuses on states that were slave states immediately prior to the Civil War; the coefficient on *black*rate* is 0.1428 and significant at the 1% level. A χ^2 test rejects the null hypothesis that the coefficients on *black*rate* are the same across the two samples at the 1% level. Columns 3 and 4 present results from splitting the sample into states with and without anti-miscegenation laws in 1967; Columns 5 and 6, results from splitting the sample into states with and without fair housing laws in 1968; and Columns 7 and 8, results from splitting the sample into states with low or high interracial bias. In each case, the coefficient on *black*rate* is larger in magnitude for black individuals residing in states with higher levels of discrimination. The null hypotheses that the coefficients for *black*rate* are the same across the two samples for the *anti-miscegenation law* state measure can be rejected at the 5% level, for the *no fair housing law* state measure at the 1% level, and for the *interracial marriage bias* state measure at the 10% level. The results in Table 7 indicate that black individuals residing in states with a history of discrimination were more likely to transition into self-employment following an increase in credit card interest rate ceilings than were black individuals in other states.

4.4. Additional robustness checks

The validity of our empirical results relies on several assumptions. First, we treat states' elimination of credit card interest rate ceilings as an exogenous shock, conditional on the control variables included in regressions. This is the strongest assumption we make in our analysis and requires additional analysis. The text of the *Marquette* decision does not mention the impact of credit cards on entrepreneurs, and in general we surmise that it is unlikely that states removed interest rate ceilings because

Table 7

Effect of state-level credit card interest rate ceilings on entrepreneurial entry, by state-level discrimination measure.

This table reports split sample results from linear probability models of transitions into self-employment on state level-credit card interest rate ceilings, black, and their interaction, using data from the *Current Population Survey* for 1971–1990. For each set of regressions, the data are split into two mutually exclusive samples: slave state in the year immediately prior to the Civil War (yes or no); anti-miscegenation law not repealed until after the US Supreme Court's 1967 decision in *Loving v. Virginia* (yes or no); no fair housing law until federally mandated by the Fair Housing Act of 1968 (yes or no); racial bias rate, as measured by the interracial marriage rate (low or high). The variable transition into self-employment equals one if the individual worked in paid employment in the previous year but switched into self-employment in the current year and zero otherwise. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year. This implies the rate ceiling varies from 24 for no-limit states prior to 1981 and 25 for no-limit states in 1981 and after. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, non-metro area indicator, local unemployment rate, and percent of local population living in rural areas. Fixed effects for years, 67 industries and 347 metropolitan statistical area (MSA)-state areas are included. Robust standard errors are included in brackets and clustered at the MSA-state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	Sample restrictions							
	Former slave state?		Anti-miscegenation law?		No fair housing law?		Interracial marriage bias	
	No (1)	Yes (2)	No (3)	Yes (4)	No (5)	Yes (6)	Low bias (7)	High bias (8)
Rate	0.0387** [0.0173]	−0.0125 [0.0442]	0.0352** [0.0171]	0.0076 [0.0514]	0.0273 [0.0183]	0.0365 [0.0333]	0.0242 [0.0178]	0.0371 [0.0351]
Black	0.0037 [0.0040]	−0.0296*** [0.0069]	0.0021 [0.0039]	−0.0263*** [0.0078]	0.0016 [0.0039]	−0.0217*** [0.0057]	−0.0026 [0.0037]	−0.0227*** [0.0078]
Black*rate	−0.0183 [0.0192]	0.1421*** [0.0367]	−0.0098 [0.0187]	0.1236*** [0.0421]	−0.0083 [0.0192]	0.0980*** [0.0295]	0.0104 [0.0180]	0.1034** [0.0414]
Individual characteristics	Y	Y	Y	Y	Y	Y	Y	Y
Industry dummies	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
MSA-state fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Trend*MSA-state fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Number of observations	386,543	160,069	389,214	157,398	305,875	240,737	343,985	202,627
R-squared	0.0149	0.0187	0.0146	0.0195	0.0149	0.0173	0.0154	0.0181

credit-constrained black or white entrepreneurs lobbied the statehouse to change the law. In fact, recent research suggests that, if anything, incumbents are more likely to engage in this type of political activity than potential entrants (Rajan and Zingales, 2003).

To more rigorously test our assumption, we run a series of state-level hazard analyses predicting when a state removes its credit card interest rate ceiling, the results of which are presented in Table 8. To conduct the hazard analysis, we first aggregate CPS data to the state level and then match the data to state-level political economy variables provided by Randall Kroszner and Philip Strahan. Column 1 includes all the demographic variables from the CPS, including *self-employed* and *black*. Column 2 adds an interaction between *black* and *self-employed*. Column 3 adds four variables that Kroszner and Strahan (1999) show affect state level adoption of bank deregulation: *small bank share of assets*, the *difference in the capital-asset ratio* between large and small banks, the *share of small firms* in the state, and an indicator equal to one if there is *single party control of state government*. Only *single party control of state government* appears to weakly predict the timing of a state's removal of its credit card interest rate ceiling. Across the columns, the coefficients on *self-employed*, *black*, and *black*self-employed* are insignificant, suggesting that, conditional on the control variables, a state's removal of its credit card interest rate ceiling is exogenous to the variables of interest in our analyses. Similar to the finding in Kroszner and Strahan

(1999), credit card deregulation, like bank deregulation, can be partially explained by political economy factors.

Second, we assume that black entrepreneurial entry following an increase in the interest rate ceiling is due to access to credit cards as opposed to some other mechanism. Summary statistics presented in Section 2 show that credit card ownership and activity increased following a state's switch to no limit. To more closely link self-employment to credit card ownership we next use data from the *Survey of Consumer Finance* to examine the effect of rate on levels of self-employment and the extent to which this effect varies by credit card ownership. In Table 9, Column 1, the coefficient on *black*rate* is positive and significant, indicating that black individuals residing in a state that removes its credit card interest rate ceiling were more likely to be self-employed. This result accords with the basic result presented in Table 4. In Columns 2 and 3 we investigate the effect of credit card ownership on self-employment. To do this, we split the sample into individuals who own a credit card in Column 2 and individuals who do not own a credit card in Column 3. The coefficient on *black*rate* is positive and significant for the subsample that owns a credit card and positive but not significant for the subsample that does not own a credit card. We interpret this set of results as weak evidence that black individuals who own a credit card are more likely to be self-employed if they live in a state with no ceiling on credit card interest rates. While consistent with our argument, the difference in coefficients across Columns 2–3 is

Table 8

Hazard models predicting when a state removes its credit card interest rate ceiling.

This table reports hazard models predicting when a state removes its credit card interest rate ceiling. We aggregate *Current Population Survey* data from 1970 to 1990 to the state level and match to state-level data provided by Randall Kroszner and Philip Strahan. Variables from Kroszner and Strahan are small bank share of assets, the difference in the capital asset ratio between large and small banks, the share of small firms in the state, and an indicator equal to one if there is single party control of the state government. Kroszner and Strahan (1999, Table III, Column 6) show that these variables affect the timing of state bank branching deregulation. Demographic variables and state and year fixed effects are included in all models. Demographic variables include female percent, average age, high school graduation rate, marriage rate, homeownership rate, average household income, average unemployment rate, and percent of population living in rural areas. Robust standard errors are included in brackets and clustered at the state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	(1)	(2)	(3)
Self-employed	−0.7819 [0.711]	−0.9225 [0.750]	−0.0919 [1.184]
Black	−0.0828 [0.199]	−0.2172 [0.243]	0.0797 [0.350]
Black*self-employed		3.4596 [2.587]	6.0627 [4.985]
Small bank share of assets			0.8093 [0.786]
Difference in small-large bank capital asset ratio			1.5156 [1.741]
Share of small firms			0.5866 [0.581]
Single party control of state government			−0.0323* [0.019]
Demographic variables	Y	Y	Y
State and year fixed effects	Y	Y	Y
Number of observations	558	558	314
R-squared	0.4	0.4	0.48
Number of clusters	51	51	37

Table 9

Effect of state-level credit card interest rate ceilings on entrepreneurship levels, using Survey of Consumer Finance (SCF) data.

This table reports the linear probability of self-employment levels on state-level credit card interest rate ceilings, black, and their interaction, using data from SCF for 1977, 1983, and 1986. The sample is split by credit card ownership in columns 2–3. Credit card interest rate ceilings for states with no limit are set equal to the highest rate ceiling across states in that year. This implies the rate ceiling varies from 24 for no-limit states prior to 1981 and 25 for no-limit states in 1981 and after. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, urban area indicator, local unemployment rate, and percent of local population living in rural areas. Fixed effects for years and 36 states are included (the SCF excludes DC, HI, ID, KS, MD, MT, ND, NH, NM, NV, RI, VT, WV, and WY). Robust standard errors are included in brackets and clustered at the state level.

*Significant at 10%; **Significant at 5%; ***Significant at 1%.

	(1)	Owns credit card?	
		No (2)	Yes (3)
Rate	−0.2146 [0.2478]	0.007 [0.4017]	−0.2339 [0.3071]
Black	−0.1462*** [0.0291]	−0.0983 [0.0592]	−0.1347*** [0.0345]
Black*rate	0.6147*** [0.1323]	0.3055 [0.3015]	0.5830*** [0.1501]
Individual characteristics	Y	Y	Y
Year fixed effects	Y	Y	Y
State fixed effects	Y	Y	Y
Number of observations	4889	1203	3686
R-squared	0.029	0.049	0.034

not statistically significant. The low statistical power of the test is not surprising, however, given the low number of observations in the SCF data-set.

Our analysis relies on several additional assumptions. We assume that within-state changes to credit card interest rate ceilings had an immediate effect on the rates offered to individuals with credit cards in that state and that rate ceilings in other states had little to no effect on the rates offered within state. Knittel and Stango (2003) provide evidence supportive of this assumption. The differences-in-differences research design compares changes in states that switch to no limit to changes in states that do not. This assumption means that any effect we find could be attenuated from the actual effect. For example, while a state could have retained an 18% ceiling on credit card interest rates, individuals in that state could, in later periods, be using out-of-state credit cards with much higher interest rates issued by a bank in a no-limit state. Hence, any difference in self-employment or credit card use between such a state and a state that changes from an 18% ceiling to no ceiling is reduced. The direction of this bias works against us finding a result. We also assume that the types of credit cards offered to individuals in no-limit states were similar to the types of credit cards offered to individuals in limit states. This assumption accords well with historical features of the credit card industry. Prior to the 1990s, most cards were offered with a fixed rate not pegged to any market rate, frequent flyer plans and other inducements were uncommon, and the cards were more or less homogeneous (Stango, 2000; Knittel and Stango, 2003).

5. Discussion and conclusion

Our paper examines how financial deregulation impacts entrepreneurial activity. We use state-level variation in credit card interest rate ceilings, which were eliminated altogether in several states following the Supreme Court's 1978 *Marquette* decision, to study the differential effect of credit cards on black and white entrepreneurial entry. Prior work has demonstrated that credit card deregulation led to an increase in the probability of owning a credit card (Zinman, 2002), and we provide additional evidence that it leads to an increase in the APR on the card and an increase in the amount of credit card debt. We next examine how the increase in supply of credit cards and credit card debt affected entrepreneurial entry. We use transitions from paid employment into self-employment to measure entrepreneurial entry and show that credit card deregulation increased entrepreneurial entry, especially for black individuals. We also show that the differential effects on black entrepreneurial entry were amplified in states with a history of discrimination.

This work contributes to literature exploring the implications of financial development, regulation, and deregulation. While several studies have examined the impact of US bank deregulation on growth and firm formation (Black and Strahan, 2002; Cetorelli and Strahan, 2006; Bertrand, Schoar, and Thesmar, 2007; Huang, 2008; and Kerr and Nanda, 2009), our paper is the first to explore the impact of credit card deregulation on key economic activities, such as entrepreneurship.

We believe our empirical findings have two major implications. First, credit cards are an important means of entrepreneurial finance and, second, black entrepreneurs

faced discrimination-based barriers to entry in the 1970s and 1980s and used credit cards as a mechanism to overcome those barriers. In the first case, our findings, which are based on results from a quasi-natural experiment, provide the first robust evidence we are aware of in favor of anecdotal stories linking credit cards to entrepreneurial entry. For example, film producer Spike Lee and Google cofounders Sergey Brin and Larry Page are among the many entrepreneurs to use credit cards to fund entrepreneurial ventures, but no large-scale empirical studies have assessed the economic significance of this phenomenon (Scott, 2009; McGarvey, 2000). In addition, these findings could be especially important when assessing the impact of the 2008 financial crisis on access to credit for entrepreneurs and small businesses. While lending to small businesses and available credit lines declined precipitously during the crisis (Council of Economic Advisers, 2011, Chapter 7), further analysis is required to assess whether these developments could have disproportionately harmed particular groups of entrepreneurs.

The second implication of our findings—that the differential effect of credit card deregulation on black entrepreneurs could be attributable to discrimination-based barriers to entry—accords well with existing empirical evidence on discrimination-based frictions in lending markets (Fairlie and Robb, 2008; Ravina, 2008; Pope and Sydnor, 2011). As suggested by Blanchflower, Levine, and Zimmerman (2003), black entrepreneurs could be more likely to use credit cards than white entrepreneurs to circumvent discrimination in lending. While our study focuses on a specific time period, 1971–1990, recent research (Ravina, 2008; Cohen-Cole (2011); Pope and Sydnor, 2011) demonstrates that discrimination still

Table A1

Effect of state-level credit card interest rate ceilings on entrepreneurial entry with various rate definitions.

This table reports linear probability models of transitions into self-employment on state-level credit card interest rate ceilings, black, and their interaction, using data from the *Current Population Survey* for 1971–1990. The variable *transition into self-employment* equals one if the individual worked in paid employment in the previous year but switched into self-employment in the current year and zero otherwise. In Column 1, credit card interest rate caps for states with no limit are set equal to the highest rate cap across states in all years. This implies the rate cap is 25 for no-limit states in all years. In Column 2, credit card interest rate caps for states with no limit are set equal to 25 across all years. In Column 3, credit card interest rate caps for states with no limit are set equal to 30 across all years. In Column 4, a dummy variable equal to one is used to indicate states with no limit on credit card interest rates and zero otherwise. Individual characteristics include female, age, age squared, high school graduate, married, homeowner, household income, non-metro area indicator, local unemployment rate, and percent of local population living in rural areas. Fixed effects for years, 67 industries and 347 metropolitan statistical area (MSA)-state areas are included. Robust standard errors are included in brackets and clustered at the MSA-state level. *Significant at 10%; **Significant at 5%; ***Significant at 1%.

	Original model (1)	Rate for no-limit states=0.25 (2)	Rate for no-limit states=0.30 (3)	Dummy: no limit=1; limit=0 (4)
Rate	0.0300** [0.0125]	0.0297** [0.0119]	0.0232*** [0.0087]	0.0020** [0.0010]
Black	−0.0087** [0.0035]	−0.0089*** [0.0034]	−0.0084*** [0.0027]	−0.0020*** [0.0005]
Black*rate	0.0362** [0.0177]	0.0389** [0.0165]	0.0339*** [0.0130]	0.0048*** [0.0017]
Individual characteristics	Y	Y	Y	Y
Industry dummies	Y	Y	Y	Y
Year fixed effects (1971–1990)	Y	Y	Y	Y
MSA-state fixed effects	Y	Y	Y	Y
Trend*MSA-state fixed effects	Y	Y	Y	Y
Number of observations	546,612	546,612	546,612	546,612
R-squared	0.0158	0.0158	0.0159	0.0158
Number of clusters	347	347	347	347

Table A2

Comparison of risk sets for individuals transitioning into self-employment.

This table reports selected summary statistics using data from the *Current Population Survey* for 1971–1990 across workers in two sectors: those in the paid employment at $t-1$ and those in unemployment at $t-1$.

	Status at $t-1$				
	Paid employment		Unemployment		T -test for differences
	$(N=562043)$		$(N=28745)$		
	(1)		(2)		
	Mean	Standard deviation	Mean	Standard Deviation	T -statistic
Black	0.086	0.281	0.184	0.387	-56.20
Female	0.372	0.483	0.647	0.478	-94.06
Homeowner	0.544	0.498	0.463	0.499	26.95
Household income	29478	29790	17672	28324	65.69
Age	37.072	12.227	33.185	12.549	52.50
High school grad	0.784	0.412	0.678	0.467	42.05
Married	0.660	0.474	0.513	0.500	51.11

affects black borrowers. Hence, it could still be the case that credit cards are an important mechanism for overcoming discrimination-based barriers to entry and would suggest that policy makers consider the differential effects that policies could have across a heterogeneous population of entrepreneurs and firms.

Appendix

This appendix contains tables with additional results described in the body of the manuscript.

Robustness tests with different rates

Table A1 replicates the results from Table 4, Column 1 using different approaches to account for the interest rate ceiling when a state eliminates ceilings altogether. Column 1 replicates the main results from Table 4. In Column 2 we use a rate ceiling of 25% for no-limit states across all years; in Column 3 we use a rate ceiling of 30% for no-limit states across all years; in Column 4 we use a dummy variable equal to one when the state has no limit and zero otherwise. The coefficient on *black*rate* is positive and significant at the 5% level or better in all cases.

Comparison of risk sets

Table A2 presents summary statistics across workers in two sectors: those in paid employment at $t-1$, and those in unemployment at $t-1$. *T*-tests reveal the groups differ along a number of dimensions. Individuals who are unemployed instead of in paid employment are significantly more likely to be black, female, non-homeowners, younger, and non-high school grads, are less likely to be married, and have lower household income. The difference across these two population groups underscores our focus on the transition from paid employment to self-

employment, instead of focusing on transitions from unemployment to self-employment in results subsequent to those presented in Table 4.

References

- Acemoglu, D., Johnson, S., Robinson, J.A., 2001. The colonial origins of comparative development: an empirical investigation. *American Economic Review* 91 (5), 1369–1401.
- Ausubel, L.M., 1997. Credit card defaults, credit card profits, and bankruptcy. *American Bankruptcy Law Journal* 71 (Spring), 249–270.
- Beck, T., Levine, R., Levkov, A., 2010. Big bad banks: the impact of US branch deregulation on income distribution. *Journal of Finance* 65 (5), 1637–1667.
- Benmelech, E., Moskowitz, T., 2010. The political economy of financial regulation: evidence from US state usury laws in the 19th century. *Journal of Finance* 65 (3), 1029–1073.
- Bertrand, M., Duflo, E., Mullainathan, S., 2007. How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics* 119 (1), 249–275.
- Bertrand, M., Schoar, A., Thesmar, D., 2007. Banking deregulation and industry structure: evidence from the French banking reforms of 1985. *Journal of Finance* 62 (2), 597–628.
- Besley, T., Burgess, R., 2004. Can labor market regulation hinder economic performance? Evidence from India. *Quarterly Journal of Economics* 113 (1), 91–134.
- Black, S.E., Strahan, P.E., 2002. Entrepreneurship and bank credit availability. *Journal of Finance* 57 (6), 2807–2833.
- Blanchflower, D.G., Levine, P.B., Zimmerman, D.J., 2003. Discrimination in the small business credit market. *Review of Economics and Statistics* 85 (4), 930–943.
- Capon, N., 1982. Credit scoring systems: a critical approach. *Journal of Marketing* 46 (2), 82–91.
- Cetorelli, N., Strahan, P.E., 2006. Finance as a barrier to entry: bank competition and industry structure in local US markets. *Journal of Finance* 61 (1), 437–461.
- Cohen-Cole, E., 2011. Credit card redlining. *Review of Economics and Statistics* 93 (2), 700–713.
- Collins, W.J., 2004. The housing market impact of state-level anti-discrimination laws, 1960–1970. *Journal of Urban Economics* 55, 534–564.
- Council of Economic Advisers, 2011. *Economic Report of the President*. Government Printing Office, Washington, DC.
- DeMuth, C.C., 1986. The case against credit card interest rate regulation. *Yale Journal on Regulation* 3 (1), 201–242.
- Evans, D., Jovanovic, B., 1989. An estimated model of entrepreneurial choice under liquidity constraints. *Journal of Political Economy* 97, 808–827.
- Fairlie, R.W., 1999. The absence of the African American-owned business: an analysis of the dynamics of self-employment. *Journal of Labor Economics* 17 (1), 80–108.
- Fairlie, R.W., Robb, A.M., 2008. *Race and Entrepreneurial Success: Black-, Asian-, and White-Owned Businesses in the United States*. MIT Press, Cambridge, MA.
- Fazzari, S., Hubbard, G., Peterson, B., 1988. Financing constraints and corporate investment. *Brookings Papers on Economic Activity* 19, 141–195.
- Fryer Jr., R.G., 2007. Guess who's been coming to dinner? Trends in interracial marriage over the 20th century. *Journal of Economic Perspectives* 21 (2), 71–90.
- Furletti, M., 2004. The debate over the National Bank Act and the preemption of state efforts to regulate credit cards. *Temple Law Review* 77, 425.
- Garmaise, M., Moskowitz, T.J., 2006. Bank mergers and crime: the real and social effects of credit market competition. *Journal of Finance* 61 (2), 495–538.
- Gross, D., Souleles, N.S., 2002. Do liquidity constraints and interest rates matter for consumer behavior? Evidence from credit card data. *Quarterly Journal of Economics* 117 (1), 149–185.
- Hellmann, T., Lindsey, L., Puri, M., 2008. Building relationships early: banks in venture capital. *Review of Financial Studies* 21 (2), 513–541.
- Hochberg, Y., Ljungqvist, A., Lu, Y., 2007. Whom you know matters: venture capital networks and investment performance. *Journal of Finance* 62, 251–301.
- Holtz-Eakin, D., Joulfaian, D., Rosen, H., 1994. Entrepreneurial decisions and liquidity constraints. *RAND Journal of Economics* 23, 334–347.
- Hsu, D., 2004. What do entrepreneurs pay for venture capital affiliation? *Journal of Finance* 59, 1805–1844.

- Huang, R., 2008. Evaluating the real effect of bank branching deregulation: comparing contiguous counties across US state borders. *Journal of Financial Economics* 87, 678–705.
- Hurst, E., Lusardi, A., 2004. Liquidity constraints, household wealth, and entrepreneurship. *Journal of Political Economy* 112 (21), 319–347.
- Kaplan, S., Zingales, L., 1997. Do investment-cash flow sensitivities provide useful measure of financing constraints? *Quarterly Journal of Economics* 112, 169–215.
- Kerr, W.R., Nanda, R., 2009. Democratizing entry: banking deregulation, financing constraints, and entrepreneurship. *Journal of Financial Economics* 94 (1), 124–149.
- Kortum, S., Lerner, J., 2000. Assessing the contribution of venture capital to innovation. *RAND Journal of Economics* 31, 674–692.
- Knittel, C.R., Stango, V., 2003. Price ceilings as focal points for tacit collusion: evidence from credit cards. *American Economic Review* 93 (5), 1703–1729.
- Kroszner, R.S., Strahan, P.E., 1999. What drives deregulation? Economics and politics of the relaxation of bank branching restrictions. *Quarterly Journal of Economics* 114 (4), 1437–1467.
- Levine, R., 2005. Finance and growth: theory and evidence. In: Aghion, P., Durlauf, S.N. (Eds.), *Handbook of Economic Growth*, North Holland, Amsterdam, pp. 865–934.
- Levine, R., Levkov, A., Rubinstein, Y., 2008. Racial discrimination and competition. Unpublished working paper 14273. National Bureau of Economic Research, Cambridge, MA.
- McGarvey, R., 2000. Search us, says Google. *Technology Review* (November), <<http://www.technologyreview.com/web/12219/>>.
- Morse, A., 2011. Payday lenders: heroes or villains? *Journal of Financial Economics* 102 (1), 28–44.
- Petersen, M.A., Rajan, R.G., 2002. Does distance still matter? The information revolution in small business lending. *Journal of Finance* 57 (6), 2533–2570.
- Pope, D.G., Sydnor, J.R., 2011. What's in a picture? Evidence of discrimination from prosper.com. *Journal of Human Resources* 46 (1), 53–92.
- Puri, M., Robinson, D.T., 2009. The economic psychology of entrepreneurship and family business. Duke University working paper.
- Rajan, R.G., Zingales, L., 2003. The great reversals: the politics of financial development in the twentieth century. *Journal of Financial Economics* 69 (1), 5–50.
- Ravina, E., 2008. Love and loans: the effect of beauty and personal characteristics in credit markets. Unpublished working paper. New York University, New York.
- Robb, A.M., Fairlie, R.W., Robinson, D.T., 2009. Financial capital injections among new black and white business ventures: evidence from the Kauffman Firm Survey. Unpublished working paper. Duke University, Durham, NC.
- Scott, R., 2009. The use of credit card debt by new firms. White paper, Kauffman Foundation, <http://www.kauffman.org/uploadedFiles/kfs_credit_card_debt_report.pdf>.
- Scott, R., 2010. Plastic capital: credit card debt and new business survival. Unpublished working paper. Monmouth University, West Long Beach, NJ.
- Stango, V., 2000. Competition and pricing in the credit card market. *Review of Economics and Statistics* 82 (3), 499–508.
- The Cost of Personal Borrowing in the United States, various years. South Bend, IN: Financial Publishing Company.
- Time, 1986. Mounting doubts about debts. Barbara Rudolph, March 31.
- Wolters, J., 2006. Did unilateral divorce raise divorce rates? A reconciliation and new results. *American Economic Review* 96 (5), 1802–1820.
- Zarutskie, R., 2006. Evidence on the effects of bank competition on firm borrowing and investment. *Journal of Financial Economics* 81, 503–537.
- Zinman, J., 2002. Liquidity and consumer behavior: some evidence from the deregulation of credit card interest rate ceilings. Unpublished working paper. Dartmouth College, Hanover, NH.